United States Patent	12403988
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Yim; Ki Ho et al.

Corner structure and liquefied gas storage tank having same

Abstract

Disclosed is a corner structure (**100**) of a liquefied gas storage tank, the corner structure (**100**) being installed on a corner of a liquefied gas-carrying storage tank to support sealing walls (**51, 52**) that prevent the liquefied gas from leaking. The corner structure (**100**) comprises: two insulation members (**110**) arranged, oriented in different directions, on the inner surface of structural walls of a ship; and operational members (**130**) which are disposed on respective insulation members (**110**), and to which the support sealing walls (**51, 52**) adhere. The operational members (**130**) may be attached to the insulation members (**110**) so as to allow sliding against same.

Inventors: **Yim; Ki Ho (Seoul, KR), Cho; Yong Bum (Seoul, KR), Yang; Young Chul (Gyeonggi-do, KR), Seo; Heung Seok (Gyeonggi-do, KR), Lee; Yeongbeom (Incheon, KR), Yoon; Youngkeun (Incheon, KR), Han; Hae Chul (Incheon, KR), Oh; Byungteak (Incheon, KR), Yun; Eunyoung (Incheon, KR)**

Applicant: **KOREA GAS CORPORATION** (Daegu, KR)

Family ID: **80253933**

Assignee: **KOREA GAS CORPORATION (Daegu, KR)**

Appl. No.: **18/036890**

Filed (or PCT Filed): **January 06, 2021**

PCT No.: **PCT/KR2021/000114**

PCT Pub. No.: **WO2022/119045**

PCT Pub. Date: June 09, 2022

Prior Publication Data

Document Identifier	Publication Date

Foreign Application Priority Data

KR	10-2020-0167214	Dec. 03, 2020

Publication Classification

Int. Cl.: **F17C3/02** (20060101); **B63B25/16** (20060101); **B65D90/04** (20060101)

U.S. Cl.:

CPC **B63B25/16** (20130101); **B65D90/04** (20130101); **F17C3/027** (20130101); F17C2201/052 (20130101); F17C2203/0358 (20130101); F17C2221/033 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6374761	12/2001	Dhellemmes	220/901	F17C 13/001
7717288	12/2009	Yang	220/560.07	B63B 25/14
11674643	12/2022	Sassi	141/1	F17C 3/04
2017/0159888	12/2016	Han	N/A	B63B 25/16

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2002181288	12/2001	JP	N/A
1006493170000	12/2005	KR	N/A
1020120013247	12/2011	KR	N/A
1020120039861	12/2011	KR	N/A
101669532	12/2015	KR	N/A
101865167	12/2017	KR	N/A
1021295610000	12/2019	KR	N/A
WO-2019012237	12/2018	WO	F17C 3/027
WO-2019215404	12/2018	WO	F17C 3/027

OTHER PUBLICATIONS

“English translation of Office Action of Korea Counterpart Application”, issued on Nov. 25, 2021, pp. 1-5. cited by applicant
“Search Report of Europe Counterpart Application”, issued on Sep. 13, 2024, pp. 1-7. cited by applicant

Primary Examiner: Impink; Mollie

Attorney, Agent or Firm: JCIPRNET

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a corner structure of a liquefied gas storage tank, and more particularly, to a corner structure arranged to install a sealing wall at a corner portion of a liquefied gas storage tank for storing liquefied gas, which is a liquid in a cryogenic state.

BACKGROUND ART

(2) In general, liquefied gas includes liquefied natural gas (LNG), liquefied petroleum gas (LPG), liquefied ethane gas, liquefied ethylene gas, liquefied nitrogen, liquefied carbon dioxide, liquefied ammonia, and the like.

(3) For example, LNG is liquefied natural gas which is one of a fossil fuel, and LNG storage tanks are classified into onshore storage tanks installed on the ground or buried in the ground and mobile storage tanks installed in transportation vehicles, such as cars and vessels, depending on locations where the LNG storage tanks are installed.

(4) Liquefied gases, such as LNG and LPG described above have a risk of explosion when exposed to impact and are stored in a cryogenic state, and thus, storage tanks for storing LNG or LPG have a structure in which impact resistance and fluid-tightness are firmly maintained.

(5) Also, compared to onshore storage tanks with little mobility, liquefied gas storage tanks installed in vehicles and ships with mobility should take measures against mechanical stress caused by mobility. However, since a liquefied gas storage tank installed in a vessel equipped with countermeasures against mechanical stress may also be used for in an onshore storage tank, a structure of a liquefied gas storage tank installed in a vessel will be described as an example in this specification.

(6) A vessel in which a storage tank for a liquefied gas, such as LNG, is installed usually has a dual structure hull including an outer wall forming the exterior and an inner wall formed inside the outer wall. The inner wall and the outer wall of the vessel may be connected by a connecting wall to be integrated, and in some cases, vessels may include a hull having a unitary structure without the inner wall.

(7) Also, the inside of the hull, i.e., the inside of the inner wall, may be divided by one or more bulkheads. The bulkhead may be formed by a known cofferdam installed in a typical LNG carrier or the like.

(8) Each of internal spaces divided by the bulkhead may be utilized as a storage tank storing cryogenic liquid, such as LNG.

(9) Here, an inner circumferential wall surface of the storage tank is sealed in a fluid-tight state by a sealing wall. That is, the sealing wall forms one storage space by integrally connecting a plurality of metal plates to each other by welding, and accordingly, the storage tank may store and transport LNG without leakage.

(10) This sealing wall is connected to the inner wall or bulkhead of the vessel by a plurality of anchor structures. Therefore, the sealing wall cannot be moved relative to the hull.

(11) An insulating wall is disposed between the sealing wall and the inner wall or bulkhead to form an insulating layer. The insulating wall may include a corner structure disposed at a corner portion of the storage tank, an anchor structure disposed around the anchor member, and a planar structure disposed at a flat portion of the storage tank. That is, the overall insulating layer may be formed in the storage tank by the corner structure, the anchor structure, and the planar structure.

(12) Here, the anchor structure includes an anchor member directly connecting and fixing the hull to the sealing wall and an insulating member installed around the anchor member.

(13) In addition, the sealing wall is mainly supported by the anchor structure, and the planar structure only supports a load of LNG applied to the sealing wall, and there is no direct coupling between the planar structure and the anchor structure.
(14) FIG. **1** is a cross-sectional view illustrating a portion of a corner of an LNG storage tank according to the related art.
(15) In a related art LNG storage tank **10** shown in FIG. **1**, secondary insulating walls **22**, **32**, and **42** and primary insulating walls **24**, **34**, and **44** are sequentially installed on an inner wall **12** or a bulkhead **14**, which is a hull structure, to insulate the inside and outside of the storage tank. In addition, secondary sealing walls **23**, **33**, and **43** are installed between the secondary insulating walls **22**, **32**, **42** and the primary insulating walls **24**, **34**, and **44**, and a primary sealing wall **50** is installed on surfaces of the primary insulating walls **24**, **34**, and **44** to seal the inside and outside of the storage tank doubly.
(16) The LNG storage tank **10** configured as described above includes a corner structure **20** installed at an inner corner portion, an anchor structure **30** installed at regular intervals on a bottom surface, and a planar structure **40** disposed between the corner structure **20** and the anchor structure **30** or between the anchor structure **30** and the anchor structure **30** and slidably movable. Here, the corner structure **20**, the anchor structure **30**, and the planar structure **40** may be prefabricated as respective unit modules and then assembled to the storage tank **10**, and the primary sealing wall **50** may be installed thereon to fluid-tightly sealing the insulating wall, thereby providing a space in which the LNG may be stored therein.
(17) As shown in FIG. **1**, the corner structure **20**, the anchor structure **30**, and the planar structure **40** may include primary insulating walls **24**, **34**, and **44**, secondary insulating walls **22**, **32**, and **42**, and secondary sealing walls **23**, **33**, and **43**.
(18) Meanwhile, in each of the structures **20**, **30**, and **40**, a secondary sealing wall of each unit module and a contact surface of each insulating wall may be bonded to each other by an adhesive to be integrally formed. Typically, the secondary insulating walls **22**, **32**, and **42** include polyurethane foam, which is an insulation material, and a plate material adhered to a lower portion thereof. And, the primary insulating walls **24**, **34**, **44** are formed of polyurethane foam and a plate material adhered thereto by an adhesive. In addition, the primary sealing wall is installed on top of the primary insulating walls **24**, **34**, **44** and fixed to the anchor structure **30** by welding.
(19) In addition, a flange **42***a* larger than the secondary insulating wall **42** is formed at a lower end of the secondary insulating wall **42** of the planar structure **40**. The flange **42***a* is inserted into a recess formed at a lower end of the anchor structure **30** and is installed to be slidably movable.
(20) In the illustrated example, each anchor structure **30** has an anchor support rod **36**, a fixing member **37** located at a lower portion, an anchor secondary insulating wall **32**, and an anchor primary insulating wall **34**, and a secondary sealing wall **33** is connected between the anchor secondary insulating wall **32** and the anchor primary insulating wall **34**. One end of the anchor support rod **36** is connected to the primary sealing wall **50** and the other end thereof is connected to a hull inner wall **12** by the fixing member **37**.
(21) Meanwhile, in the anchor structure **30**, the primary sealing wall **50** is welded and coupled to the upper end of the anchor support rod **36**.
(22) In addition, the anchor structure **30** is located at a connection point of adjacent planar structures **40** to connect them, and the planar structure **40** is fixed to the hull inner wall **12** or the bulkhead **14** forming the storage tank **10**. In addition, the fixing member **37** of the anchor structure **30** is installed around the anchor support rod **36**.
(23) However, in the related art LNG storage tank, a configuration of the insulating wall structure includes primary and secondary insulating walls and a secondary sealing wall interposed therebetween, which is complicated. In addition, the structure for connecting the secondary sealing walls of each unit module to each other is complicated, and connection work is not easy. In addition, since the structure and installation work of a connection portion of an anchor portion or

the secondary sealing wall are difficult, reliability of LNG sealing to the secondary sealing wall may be lowered, to cause leakage of LNG.

(24) In addition, the related art corner structure **20**, in which only a load of LNG applied to the sealing wall **50** is supported and the sealing wall **50** is not attached, there may be room for improvement in absorbing stress occurring during deformation of the hull or thermal deformation of the storage tank due to loading and unloading of LNG in a cryogenic state.

(25) In recent years, as engine performance has improved, the consumption of boil-off gas has decreased, and demand for a lower boiler-off rate (BOR) has gradually increased. To this end, an increase in thickness of an insulating structure to increase the insulation performance may increase a weight and an increase in the amount of shrinkage of the insulating structure for sloshing impact, causing a problem in that a relative displacement between the sealing wall and the anchor structure further increases. For this reason, the reliability of LNG sealing in the sealing wall may be lowered to cause LNG leakage.

(26) Therefore, it is necessary to continuously make efforts to improve work efficiency and reduce construction period and costs when manufacturing a storage tank by reducing the weight of each unit module, while maintaining the insulation performance of the insulating structure.

DISCLOSURE

Technical Problem

(27) The present disclosure provides a corner structure of a liquefied gas storage tank having an improved structure, capable of simplifying a structure of an insulating wall and a sealing wall and a coupling structure thereof in the liquefied gas storage tank, improving work to be easy, increasing reliability of sealing, shortening a dry time of the tank by simplifying an assembly structure and a manufacturing process, and allowing a corner portion to more efficiently resolve mechanical stress occurring in the storage tank.

Technical Solution

(28) According to an embodiment of the present disclosure, a corner structure of a liquefied gas storage tank installed at a corner of a storage tank for loading liquefied gas and supporting a sealing wall preventing leakage of liquefied gas, includes: two insulating members disposed on an inner surface of a hull structure wall to be oriented in different directions; and a movable member installed on each of the insulating members and to which the sealing wall is attached, wherein the movable member is coupled to be slidably displaced with respect to the insulating member.

(29) The insulating member may include a lower plate, a middle plate, and an upper plate having a flat plate shape; a lower heat insulator interposed between the lower plate and the middle plate; and an upper insulator interposed between the middle plate and the upper plate, wherein the upper insulator and the lower insulator are formed of an insulator of the same material.

(30) The lower insulator may have a density lower than a density of the upper insulator.

(31) The insulating member may include one or more reinforcing plates connecting the lower plate and the middle plate in the lower insulator to reinforce the lower insulator.

(32) The reinforcing plates may be arranged parallel to each other within the lower insulator, a mastic is interposed between the insulating member and the hull structure wall, and the mastic is located on a straight line with the reinforcing plate.

(33) The sealing wall may include a primary membrane and a secondary membrane, the movable member includes a primary joint portion to which the primary membrane is attached, a secondary joint portion formed to have a step difference from the primary joint portion, to which the secondary membrane is attached, and a flange portion extending from the secondary joint portion for coupling with the insulating member, wherein the flange portion is slidably interposed between upper plates of the insulating member formed of two sheets of plywood, so that the insulating member and the movable member may be combined to be relatively slidably displaceable.

(34) The secondary joint portion and the flange portion may be formed by bending a sheet of metal, and the primary joint portion is formed by adhering a metal rod having a rectangular cross-section

to the secondary joint portion.
(35) The corner structure of the liquefied gas storage tank may further include a middle insulator disposed in a space surrounded by two insulating members oriented in different directions and the hull structure wall.
(36) The corner structure of the liquefied gas storage tank may further include: a curved member disposed between upper plates of the two insulating members to support the sealing wall and having a curved surface facing an inside of the storage tank.
(37) The insulating member may include two upper plates, the movable member includes a joint portion to which the sealing wall is bonded and a flange portion extending from the joint portion, among the two upper plates, a first upper plate located on a lower side may include a concave portion in which the flange portion is seated, and a second upper plate located above the first upper plate may include an opening through which the joint portion passes, and the flange portion may be interposed between the first upper plate and the second upper plate in the concave portion.
(38) A length and width of the concave portion may be greater than a length and width of the flange portion, and a length and width of the opening portion are greater than a length and width of the joint portion.
(39) According to an embodiment of the present disclosure, there is provided a liquefied gas storage tank including a corner structure installed at a corner to support a sealing wall preventing leakage of liquefied gas, wherein the corner structure includes: two insulating members disposed on an inner surface of a hull structure wall to be oriented in different directions; and a movable member installed on each of the insulating members and to which the sealing wall is attached, wherein the movable member is coupled to be slidably displaced with respect to the insulating member.
(40) A planar structure may be disposed around the corner structure, the planar structure may include a secondary insulating panel installed on the hull structure wall and a primary insulating panel adhered to the secondary insulating panel to be adjacent to the sealing wall, the primary insulator included in the primary insulating panel and the secondary insulator included in the secondary insulating panel may be formed of an insulator of the same material, the secondary insulator has a density lower than a density of the primary insulator, and the secondary insulating panel may include one or more secondary reinforcing plates for reinforcing the secondary insulator in the secondary insulator.
(41) The sealing wall may include a primary membrane in direct contact with liquefied gas and a secondary membrane installed to be spaced apart from the primary membrane by a predetermined distance, and a support plate may be interposed between the primary membrane and the secondary membrane to maintain a constant interval therebetween.
Advantageous Effects
(42) As described above, according to the present disclosure, a corner structure of a liquefied gas storage tank having an improved structure, capable of simplifying a structure of an insulating wall and a sealing wall and a coupling structure thereof in the liquefied gas storage tank, improving work to be easy, increasing reliability of sealing, shortening a dry time of the tank by simplifying an assembly structure and a manufacturing process, and allowing a corner portion to more efficiently resolve mechanical stress occurring in the storage tank may be provided.

Description

DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a cross-sectional view illustrating a portion of a storage tank for LNG according to the related art;
(2) FIG. **2** is a perspective view of a corner structure according to an embodiment of the present

disclosure, illustrating primary and secondary membranes and a portion of a planar structure together;
(3) FIG. **3** is a cross-sectional view taken along plane A-A plane;
(4) FIGS. **4** to **8** are cross-sectional views illustrating an assembly process of a corner structure according to an embodiment of the present disclosure;
(5) FIG. **9** is a partially enlarged plan view of a corner structure according to an embodiment of the present invention; and
(6) FIG. **10** is a cross-sectional view of main portions illustrating a state in which primary and secondary membranes are bonded to a corner structure according to an embodiment of the present invention.

BEST MODE

(7) Hereinafter, configuration and operation according to an embodiment of the present disclosure will be described in detail with reference to the drawings. In addition, the following embodiments may be modified into various other forms, and the scope of the present disclosure is not limited to the following embodiments.
(8) In this specification, the expressions ‘upper portion’ and ‘lower portion’ are based on each corner structure or planar structure before being adhered to a structure wall of a hull to form a storage tank and are not based on the entire storage tank. Each corner structure or planar structure may be adhered not only to the bottom of the storage tank, but also to a ceiling and side walls. For example, when each corner structure or planar structure is adhered to the bottom of the storage tank, the ‘upper portion’ and ‘lower portion’ in each corner structure or planar structure have the same orientation as ‘upper portion’ and ‘lower portion’ in the entire storage tank, but when each corner structure or planar structure is adhered to the ceiling or side surface of the storage tank, the ‘upper portion and the ‘lower portion’ of each corner structure or planar structure have different orientations from the ‘upper portion and ‘lower portion’ of the entire storage tank.
(9) A liquefied gas storage tank formed by a corner structure **100** and a planar structure **300** according to an embodiment of the present disclosure includes an insulating wall and a sealing wall laminated the structure wall (hull; **12**, **14**) of a hull, like the storage tank described above with reference to FIG. **1**. However, compared to the related art storage tank shown in FIG. **1** in which a secondary insulating wall, a secondary sealing wall, a primary insulating wall, and a primary sealing wall are sequentially alternately laminated, in the storage tank including the corner structure **100** and the planar structure **300** according to an embodiment of the present disclosure, a sealing wall is installed on an insulating wall and the sealing wall is not interposed between insulating walls. The insulating wall may be formed by arranging a plurality of modularized insulating structures (e.g., the corner structure **100**, the planar structure **300**, etc.) on the structure walls **12** and **14** of the hull.
(10) FIG. **2** is a perspective view of a corner structure according to an embodiment of the present disclosure, and FIG. **3** is a cross-sectional view taken along plane A-A of FIG. **2**. FIG. **2** illustrates the corner structure **100** according to an embodiment of the present disclosure together with primary and secondary membranes **51** and **52** and a portion of a planar structure **300**. The shapes of the first and secondary membranes **51** and **52** and the shape of the planar structure **300** are not limited to those illustrated.
(11) As shown in FIGS. **2** and **3**, the corner structure **100** according to an embodiment of the present disclosure includes an insulating member **110** disposed on a surface of a wall body partitioning an internal space of the hull so that the storage tank **10** (refer to FIG. **1**) may be installed, that is, a hull structure wall, such as an inner wall **12** (refer to FIG. **1**) or the blockhead **14** (refer to FIG. **1**), and a movable member **130** supported on the insulating member **110** and to which membranes **51** and **51** for sealing are attached.
(12) Here, the movable member **130**, as will be described later, is installed to be finely displaceable with respect to the insulating member **110** when thermal deformation resulting from a change in

temperature according to loading or unloading of liquefied natural gas (LNG) in a cryogenic state or deformation of the hull due to waves occurs. That is, the movable member **130** and the insulating member **110** are configured to be relatively displaceable with respect to each other.
(13) According to an embodiment of the present disclosure, the insulating member **110** has a coupling structure with the movable member **130**, but may be configured not to have a coupling structure with the hull structure walls **12** and **14**. As will be described later, the insulating member **110** is only placed on the hull structure walls **12** and **14** with a mastic **18** interposed therebetween, and may not be coupled by a separate mechanical coupling structure.
(14) Each insulating member **110** may be formed of, for example, a polyurethane foam insulator and plywood. However, the present disclosure is not limited by the material and structure of the insulating member **110** included in the corner structure **100**.
(15) The insulating member **110** may include a lower plate **112**, a lower insulator **114**, a middle plate **116**, an upper insulator **118**, and upper plates **122** and **124**. The lower plate **112** and the middle plate **116** may be formed of one sheet of plywood, and the upper plates **122** and **124** may be formed of two sheets of plywood.
(16) The upper insulator **118** and the lower insulator **114** may be formed of the same material, for example, polyurethane foam (PUF), and the lower insulator **114** may be foam-molded to have a density value equal to or lower than that of the upper insulator **118**. For example, the upper insulator **118** may be formed of PUF having a density of 80 to 240 kg/m 3, and the lower insulator **114** may be formed of PUF having a density of 40 to 240 kg/m 3 The upper insulator **118**, located relatively close to the cryogenic liquefied gas, is manufactured to have a relatively high density to improve insulation performance, and the lower insulator **114**, located relatively far from the cryogenic liquefied gas (located closer to the hull structure wall side), is manufactured to have a relatively low density to reduce a weight of the insulating member **110**. Therefore, it is possible to simultaneously achieve BOR improvement and weight reduction of the storage tank.
(17) In addition, the insulating member **110** may include one or more reinforcing plates **113** connecting the lower plate **112** and the middle plate **116** to reinforce the lower insulator **114** manufactured to have a relatively low density. The reinforcing plate **113** may be formed of plywood. When a plurality of reinforcing plates **113** are installed in the lower insulator **114**, the plurality of reinforcing plates **113** may be arranged in parallel with each other. Although seven reinforcing plates **113** are shown in FIG. **3**, the number of installed reinforcing plates may vary depending on a size of the insulating member **110** or a density of the lower insulator **114**. As the lower heat insulator **114**, a polyurethane foam mass formed by foam-molding may be used by cutting it to a predetermined size, that is, to a size of a space between the reinforcing plates **113**.
(18) A protective layer (not shown) formed of glass wool may be laminated on a side surface of the insulating member **110** to protect the upper insulator **118** and the lower insulator **114**.
(19) In the above, in order to reinforce the lower insulator **114** of the insulating member **110**, the use of reinforcing plates **113** arranged in parallel has been illustrated, but variations may be made such that an insulating box formed of plywood is used or the reinforcing plates are arranged in a grid form, etc. In addition, the insulating member **110** may be deformed so that the reinforcing plate for reinforcing the lower insulator **114** is not used. In addition, the insulating member **110** may be formed of a single layer of insulator, similar to the middle insulator **140** described below, instead of having a two-layer structure of an upper insulator and a lower insulator.
(20) The mastic **18** may be interposed between the insulating member **110** and the hull structure walls **12** and **14**. The mastic **18**, which is an adhesive of epoxy material, may be applied to be located on a straight line with the reinforcing plate **113**, as shown in FIG. **3**. According to the corner structure **100** of the present embodiment, only the mastic **18** may be interposed between the insulating member **110** and the hull structure walls **12** and **14**, and a fixing structure for fixing the insulating member **110** of the corner structure **100** to the hull structure walls **12** and **14**, for example, mechanical fixing members, such as stud bolts and nuts, may not be provided.

(21) The movable member **130** includes a primary joint portion **132** to which the primary membrane **51** is attached, a secondary joint portion **134** formed to have a step difference from the primary joint portion **132** and to which the secondary membrane **52** is attached, and a flange portion **136** extending from the secondary joint portion **134** for coupling with the insulating member **110**. The flange portion **136** of the movable member **130** is slidably interposed between the upper plates **122** and **124** of the insulating member **110** formed of two plywoods, so that the insulating member **110** and the movable member **130** are connected.
(22) For example, the secondary joint portion **134** and the flange portion **136** may be formed by bending a sheet of metal (for example, SUS having a thickness of 3 t), and the primary joint portion **132** may be formed by adhering a metal rod having a rectangular cross-section (for example, SUS having a thickness of 13 t) on the secondary joint portion **134**.
(23) As described above, the sealing membrane includes the primary membrane **51** forming a primary sealing wall, while directly contacting liquefied gas, and the secondary membrane **52** forming a secondary sealing wall. Each of the primary joint portion **132** and the secondary joint portion **134** may be provided in the movable member **130** so that the primary membrane **51** and the secondary membrane **52** may be joined at regular intervals, for example, by welding. A difference in height between the primary joint portion **132** and the secondary joint portion **134** may be set equal to a gap formed between the primary membrane **51** and the secondary membrane **52**.
(24) A support plate **53** may be interposed between the primary membrane **51** and the secondary membrane **52** to maintain a gap and support a load from cargo. The support plate **53** may be formed of plywood, for example.
(25) In order for the corner structure **100** shown in FIGS. **2** and **3** to be installed in a corner portion at which two wall surfaces among a plurality of wall surfaces forming the storage tank are connected at an angle of 90 degrees, two insulating members **110** are arranged to be oriented at an angle of 90 degrees. When two of the plurality of wall surfaces forming the storage tank are connected at an angle (e.g., 30 degrees, 45 degrees, 60 degrees, etc.) other than 90 degrees, the insulating member may be oriented according to the angles. In the following description and drawings, a 90-degree corner structure is described as an example, but this is only an example and the present disclosure is not limited by the angle formed by the corner structure.
(26) A space demarcated by the two insulating members **110** oriented in different directions and the hull structure walls **12** and **14** may be filled with the middle insulator **140** having a shape corresponding to the space. In FIGS. **2** and **3**, a cross-sectional shape of the middle insulator **140** is approximately square, but the shape of the middle insulator may vary according to the angle formed by the two insulating members **110**. The middle insulator **140** may be formed of PUF having a density of 40 to 240 kg/m 3, for example.
(27) A gap between the insulating member **110** and the middle insulator **140** may be filled with an insulator, such as glass wool. Glass wool may have a density less than 90 kg/m 3, for example. Glass wool may have a density of 20 to 50 kg/m 3, for example.
(28) A corner portion of the middle insulator **140**, that is, the corner portion (an upper right corner portion of the middle insulator **140** in FIG. **4**) of a portion in which the two insulating members **110** are adjacent may be chamfered to prevent damage.
(29) The insulating member may have a modified structure to be fixed on the hull structure wall in a mechanical manner, for example, by using stud bolts and nuts. In addition, the insulating member may have a modified structure so as to be fixed by the adjacent planar structure **300**.
(30) The corner structure **100** according to an embodiment of the present disclosure may further include a curved member **150** having a curved surface facing the inside of the tank. The curved member **150** may be formed of, for example, PLW or high-density polyurethane foam (for example, PUF of 80 to 240 kg/m 3). Alternatively, the curved member **150** may be formed of, for example, an organic insulator having a cell structure. The curved member **150** is disposed between the upper plates **122** and **124** of the two insulating members **110** to support the membranes **51** and

52.

(31) FIGS. **4** to **8** are cross-sectional views illustrating an assembly process of a corner structure according to an embodiment of the present disclosure, FIG. **9** is a partially enlarged plan view illustrating a partially enlarged upper plate of the movable plate of the corner structure and the insulating member, and FIG. **10** is a cross-sectional view of a main portion of the corner structure to which the primary and secondary membranes are attached.

(32) The corner structure **100** according to an embodiment of the present disclosure may be manufactured as a single module by integrally adhering the movable member **130** to the insulating member **110**. The insulating member **110** to which the movable member **130** is adhered may be manufactured at a site where a vessel having a storage tank is built or may be manufactured as a module in a nearby or remote factory and then transported to the site.

(33) As shown in FIGS. **4** and **10**, the movable member **130** may be slidably coupled to the upper plates **122** and **124** of the insulating member **110**. Specifically, among the two upper plates of the insulating member **110**, the first upper plate **122** (plywood of **15***t*) has a concave portion **122***a* in which the flange portion **136** of the movable member **130** may be seated, and an opening **124***a* into which the secondary joint portion **134** of the movable member **130** may be inserted is formed in the second upper plate **124** (plywood of **15***t*).

(34) A length and width of the concave portion **122***a* are larger than a length and width of the movable member **130**. A length and width of the opening **124***a* are greater than a length and width of the secondary joint portion **134** of the movable member **130**. As shown in FIG. **9**, gaps a and b are formed between the opening **124***a* and the secondary joint portion **134** of the movable member **130**. Further, as shown in FIG. **10**, a gap is also formed between a side wall surface of the concave portion **122***a* and the flange portion **136** of the movable member **130**.

(35) Therefore, by sequentially stacking the first upper plate **122**, the movable member **130** and the second upper plate **124** and fixing the first upper plate **122** to the second upper plate **124**, the movable member **130** may be slidably interposed between the first upper plate **122** and the second upper plate **124**.

(36) A spacer **126** may be disposed between the secondary joint portion **134** of the movable member **130** and a bottom surface of the concave portion **122***a* of the first upper plate **122**. The spacer **126** may be integrally formed with the first upper plate **122** or may be formed as a separate member. A gap is also formed between the spacer and the flange portion **136**.

(37) As shown in FIG. **5**, the curved member **150** is located between the two insulating members **110**. Both edges of the curved portion **152** of the curved member **150** are close to the movable member **130** but do not contact the movable member **130**.

(38) As illustrated in FIGS. **6** to **8**, the secondary membrane **52**, the support plate **53**, and the primary membrane **51** may be sequentially stacked on the corner structure **100** according to an embodiment of the present disclosure.

(39) The secondary membrane **52** is bonded to the secondary joint portion **134** of the movable member **130**. The secondary membrane **52** may include, for example, a secondary curved portion **52***a* that is bent at 90 degrees and a secondary flat portion **52***b* formed to have a flat plate shape. The secondary curved portion **52***a* extends between two movable members **130**, and a cross-section thereof has a substantially circular arc-shape and curved to be rounded so that the secondary curved portion **52***a* may be seated on the curved portion **152** of the curved member **150**. The secondary flat portion **52***b* may have wrinkles to respond to thermal deformation of the membrane.

(40) The support plate **53** is laminated on the secondary membrane **52**. Like the secondary membrane, the support plate **53** may include, for example, a curved portion support plate **53***a* that is bent at 90 degrees and a flat support plate **53***b* formed to have a flat plate shape. The secondary curved portion **52***a* extends between two movable members **130**, and a cross-section thereof has a substantially circular arc-shape and curved to be rounded so that the secondary curved portion **52***a* may be seated on the curved portion **152** of the curved member **150**.

(41) The primary membrane **51** is bonded to the primary joint portion **132** of the movable member **130**. Like the secondary membrane, the primary membrane **51** may include, for example, a primary curved portion **51***a*, which is a portion bent at 90 degrees, and the secondary flat portion **52***b* formed to have a flat plate shape. The primary curved portion **51***a* extends between two movable members **130**, and a cross-section thereof has a substantially circular arc-shape and curved to be rounded so that the primary curved portion **51***a* may be seated on the curved portion support plate **53***a*. The primary flat portion **51***b* may have wrinkles to respond to thermal deformation of the membrane.
(42) The support plate **53** may be interposed over the entire portion, except for a portion in which the primary and secondary membranes **51** and **52** are arranged to be parallel to each other, that is, the portion in which the wrinkles are formed, but may also be interposed partially over the remaining portion except for the portion in which wrinkles are formed.
(43) As the support plate **53**, plywood having a certain thickness may be used alone, polyurethane foam (or reinforced polyurethane foam) having a certain thickness may be used alone, or polyurethane foam (or reinforced polyurethane foam) to which plywood is adhered may be used.
(44) As described above, when loading and unloading cargo or when an external force is generated at sea, relative displacement may occur between the movable member **130** and the insulating member **110** relative to each other due to deformation of the hull or membrane. As shown in FIGS. **9** and **10**, since a size of the concave portion **122***a* formed in the first upper plate **122** of the insulating member **110** is larger than a size of the flange portion **136** of the movable member **130** and a size of the opening **124***a* formed in the second upper plate **124** of the insulating member **110** is larger than a size of the secondary joint portion **134** of the movable member **130**, even if displacement occurs, the displacement may be absorbed.
(45) In addition, when the membranes **51** and **52** shrink due to thermal deformation generated during shipment of liquefied gas, the movable member **130** to which the membranes **51** and **52** are bonded may also shrink together. At this time, both ends of the movable member **130** may be displaced while sliding finely toward the central portion of the movable member. As described above, since the flange portion **136** of the movable member **130** is slidably interposed between the first upper plate **122** and the second upper plate **124**, the coupling state of the movable member **130** to the insulating member **110** may be maintained continuously even when the movable member **130** contracts and expands.
(46) As described above, the storage tank **10** is sealed in a liquid-tight state by the first and secondary membranes **51** and **52**. That is, the storage tank **10** forms one storage space surrounded by a two-ply sealing wall by integrally connecting a plurality of metal plates to each other by welding, and accordingly, the storage tank **10** may store and transport liquefied gas without leakage.
(47) As is well known, the primary membrane **51** in direct contact with liquefied gas, such as LNG in a cryogenic state, and the secondary membrane **52** installed to be spaced apart from the primary membrane **51** have wrinkles formed to respond to changes in temperature according to loading and unloading of the liquefied gas.
(48) These primary and secondary membranes **51** and **52** may be indirectly connected to the hull structure walls **12** and **14** of the vessel through a plurality of corner structures **100** and anchor structures (not shown).
(49) Referring back to FIGS. **2** and **3**, the planar structure **300** may be arranged around the corner structure **100**. Compared to the insulation member **110** of the corner structure **100** described above, the planar structure **300** is different in that it has a structure in which a primary insulating panel **310** and a secondary insulating panel **320** are stacked.
(50) As shown in FIGS. **2** and **3**, the planar structure **300** according to an embodiment of the present disclosure for forming an insulating wall may include a primary insulating panel **310** and a secondary insulating panel **320**, and the primary insulating panel **310** and the secondary insulating

panel **320** may be bonded to each other by, for example, PU bonding to be integrated.
(51) The primary insulating panel **310** and the secondary insulating panel **320** of the planar structure **300** may be formed of, for example, an insulation material of polyurethane foam and plywood. More specifically, the primary insulating panel **310** of the planar structure **300** located closer to the sealing wall may include, for example, a primary insulator **314** formed of polyurethane foam, etc., and a primary upper plate **312** and a primary lower plate **316** bonded to the upper and lower surfaces of the primary insulator **314**, respectively. Adhesion between the primary insulator **314** and the primary upper and lower plates **312** and **316** may be achieved by, for example, PU bonding.
(52) In addition, the secondary insulating panel **320** of the planar structure **300** located to be closer to the hull structure wall may include, for example, a secondary insulator **324** formed of polyurethane foam or the like and a secondary upper plate **322** and a secondary lower plate **326** laminated on upper and lower surfaces of the secondary insulator **324**.
(53) A plurality of secondary reinforcing plates **328** may be disposed between the secondary upper plate **322** and the secondary lower plate **326** to reinforce strength. The secondary reinforcing plate **328** connecting the secondary upper plate **322** and the secondary lower plate **326** may be made of plywood. When a plurality of secondary reinforcing plates **328** are installed in the secondary insulator **324**, the plurality of secondary reinforcing plates **328** may be arranged in parallel with each other. FIG. **3** shows the planar structure **300** having ten secondary reinforcing plates **328**, but the number of reinforcing plates being installed may vary depending on the size of the planar structure **300** or the density of the secondary insulator **324**.
(54) The secondary insulator **324** may fill the space between the plurality of secondary reinforcing plates **328**. For example, the secondary insulator **324** may be cut to have a size suitable for each space and inserted thereinto.
(55) The primary insulator **314** and the secondary insulator **324** may be formed of the same material, for example, polyurethane foam, and the primary insulator **314** and the secondary insulator **324** may be foam-molded so that the density of the secondary insulator **324** is lower than that of the primary insulator **314**.
(56) According to the present disclosure, by manufacturing the planar structure **300** to have a two-layer structure by bonding the primary insulating panel **310** and the secondary insulating panel **320**, heat inflow from the outside to the inside of the storage tank may be better blocked, and by forming the secondary insulator **324** filled in the secondary insulating panel **320** to have a low density, both BOR improvement and weight reduction of the insulating structure may be achieved.
(57) Furthermore, by reinforcing the secondary insulating panel **320** by a plurality of secondary reinforcing plates **328** disposed in parallel with each other, even if an insulator having a relatively low density is used for the secondary insulating panel **320**, the strength of the secondary insulating panel **320** may be improved.
(58) The planar structure **300** formed by bonding the primary insulating panel **310** and the secondary insulating panel **320** to each other is modularized and pre-manufactured in a factory in advance, and each modular unit planar structure is transported to the site and then mounted on the hull structure wall to manufacture a storage tank.
(59) A protective layer (not shown) of a glass wool material for protecting the primary insulator **314** and the secondary insulator **324** may be laminated on the side surface of the planar structure **300**. A space between the corner structure **100** and the planar structure **300** may be filled with an insulating material, such as glass wool, or the like.
(60) In the above, in order to reinforce the secondary insulator **324** of the planar structure **300**, the use of secondary reinforcing plates **328** arranged in parallel has been described, but modification may be made such that an insulation box formed of plywood may be used or reinforcing plates may be arranged in a grid form. In addition, the planar structure **300** may be deformed so that the reinforcing plate for reinforcing the secondary insulator **324** may not be used. In addition, the

planar structure **300** may be formed of a single layer of insulator, similar to the aforementioned middle insulator **140**, instead of having a two-layer structure of a primary insulator and a secondary insulator.

(61) The mastic **18** may be interposed between the planar structure **300** and the hull structure walls **12** and **14**. The mastic **18**, which is an adhesive of epoxy material, may be applied so as to be located on a straight line with the secondary reinforcing plate **328**, as shown in FIG. **3**. The planar structure **300** according to the present embodiment may have a fixing structure for fixing the planar structure **300** to the hull structure walls **12** and **14**, for example, mechanical fixing members (not shown), such as stud bolts and nuts.

(62) An anchor unit (not shown) may be mounted at the center of the upper surface of the planar structure **300** to support the sealing wall. When the planar structure **300** includes an anchor unit, the planar structure having the anchor unit may function as an anchor structure. When manufacturing the liquefied gas storage tank, if necessary, the anchor structure and the planar structure may be properly arranged and mounted on the hull structure wall.

(63) Like the insulating member **110** of the corner structure **100**, the planar structure **300** may be modularized and pre-manufactured in a factory in advance, and each modular unit planar structure may be transported to the site, and then mounted on the hull structure wall for manufacturing a storage tank.

(64) Each of the corner structure **100**, anchor structure, and planar structure arranged in the storage tank **10** may be manufactured as one module in a separate location, and then transferred to the storage tank **10** and assembled. Due to modularization, workability may be improved when manufacturing a storage tank.

(65) The primary and secondary membranes **51** and **52** are supported by corner structure **100** and anchor structure, and the planar structure only support a load of LNG applied to the primary and secondary membranes **51** and **52**. In addition, it may be configured so that there is no direct coupling relationship between the planar structure and the corner structure **100** or between the planar structure and the anchor structure.

(66) As described above, according to an embodiment of the present disclosure, the primary membrane **51** and the secondary membrane **52** are spaced apart from each other, and only the support plate **53** is interposed therebetween and an insulator is not interposed therebetween. Since most conventional insulation barrier structures have a primary insulating wall interposed between a primary sealing wall and a secondary sealing wall in direct contact with LNG, a complicated structure is required to support the primary sealing wall by the secondary sealing wall through the primary insulating wall. In contrast, the corner structure **100** according to the present disclosure is configured not to interpose an insulator performing a separate insulating function between the primary and secondary membranes **51** and **52**, the primary and secondary membranes **51** and **52** may be relatively easily supported by the primary and secondary joint portions of the movable member **130**.

(67) In addition, according to the present disclosure, since the primary membrane **51** and the secondary membrane **52** are spaced apart from each other, even if the shape of the storage tank is deformed due to deformation of the hull due to external forces, such as waves, friction does not occur between the first and secondary membranes **51** and **52** and even if damage occurs due to an impact applied to one membrane, it is possible to prevent the damage from being directly propagated to the other membrane.

(68) Meanwhile, although the sealing is described as having a double structure by the primary and secondary membranes **51** and **52**, it is also possible to laminate three or more layers to form a multilayer structure.

(69) In addition, according to the present disclosure, the movable member **130** to which the primary and secondary membranes **51** and **52** are bonded is finely slidably connected to the insulating member **110** as described above, so that the primary and secondary membranes **51** and

52 may be stably supported with respect to the hull. Accordingly, stress caused by thermal deformation due to loading and unloading of LNG or deformation of the hull due to external forces, such as waves, may be reliably absorbed.
(70) As shown in FIG. **11**, the primary joint portion of the movable member **130** may be formed of a metal rod having a rectangular cross-section or a bent metal plate. (a) of FIG. **11** illustrates a cross-sectional view before assembly of the movable member **130** having the primary joint portion **132** formed of a metal rod, and (b) of FIG. **11** illustrates a cross-sectional view before assembly of a movable member **130**A having a primary joint portion **132**A formed of a bent metal plate.
(71) The insulator, insulating member, or insulating material used in the above embodiment of the present disclosure may include, for example, glass wool, mineral wool, polyester filler, polyurethane foam, melanin foam, polyethylene foam, polypropylene foam, silicone foam, polyvinyl chloride foam, or the like.
(72) Further, in the above embodiment of the present disclosure, it is described that the membrane is formed of, for example, corrugated stainless steel used in GTT Mark-III type, but the membrane may also be formed of, for example, Invar steel used in No. 96 of GTT.
(73) In addition, of course, the present disclosure may be equally applied to liquefied gas storage tanks installed on land as well as liquefied gas storage tanks installed inside the hull of vessels.

Claims

1. A corner structure of a liquefied gas storage tank installed at a corner of a storage tank for loading liquefied gas, the corner structure comprising: two insulating members disposed on an inner surface of a hull structure wall to be oriented in different directions; a sealing wall preventing leakage of liquefied gas; and a movable member installed on each of the insulating members and to which the sealing wall is attached, wherein the movable member is coupled to be slidably displaced with respect to each of the insulating members, the sealing wall includes a primary membrane and a secondary membrane, the movable member includes a primary joint portion to which the primary membrane is attached, a secondary joint portion formed to have a step difference from the primary joint portion, to which the secondary membrane is attached, and a flange portion extending from the secondary joint portion for coupling with each of the insulating members, the flange portion is slidably interposed between upper plates of each of the insulating members formed of two sheets of plywood, so that each of the insulating members and the movable member are combined to be relatively slidably displaceable, and the secondary joint portion and the flange portion are formed by bending a sheet of metal, and the primary joint portion is formed by adhering a metal rod having a rectangular cross-section or a U-shaped section steel formed by bending a sheet of metal to the secondary joint portion.
2. The corner structure of claim 1, wherein each of the insulating members includes a lower plate, a middle plate, and the upper plates having a flat plate shape; a lower insulator interposed between the lower plate and the middle plate; and an upper insulator interposed between the middle plate and the upper plates, wherein the upper insulator and the lower insulator are formed of an insulator of the same material.
3. The corner structure of claim 2, wherein the lower insulator has a density lower than a density of the upper insulator.
4. The corner structure of claim 2, wherein each of the insulating members includes one or more reinforcing plates connecting the lower plate and the middle plate in the lower insulator to reinforce the lower insulator.
5. The corner structure of claim 4, wherein the reinforcing plates are arranged parallel to each other within the lower insulator, a mastic is interposed between each of the insulating members and the hull structure wall, and the mastic is located on a straight line with the reinforcing plate.
6. The corner structure of claim 1, further comprising: a middle insulator disposed in a space

surrounded by two insulating members oriented in different directions and the hull structure wall.
7. The corner structure of claim 1, further comprising: a curved member disposed between upper plates of the two insulating members to support the sealing wall and having a curved surface facing an inside of the storage tank.
8. The corner structure of claim 1, wherein among the two sheets of plywood, a first plywood located on a lower side includes a concave portion in which the flange portion is seated, and a second plywood located above the firstplywood includes an opening through which the primary joint portion and the secondary joint portion pass, and the flange portion is interposed between the first plywood and the second plywood in the concave portion.
9. The corner structure of claim 8, wherein a length and width of the concave portion are greater than a length and width of the flange portion, and a length and width of the opening portion are greater than a length and width of the secondary joint portion.
10. A liquefied gas storage tank including a corner structure installed at a corner, wherein the corner structure comprises: two insulating members disposed on an inner surface of a hull structure wall to be oriented in different directions; a sealing wall preventing leakage of liquefied gas; and a movable member installed on each of the insulating members and to which the sealing wall is attached, wherein the movable member is coupled to be slidably displaced with respect to each of the insulating members, the sealing wall includes a primary membrane and a secondary membrane, the movable member includes a primary joint portion to which the primary membrane is attached, a secondary joint portion formed to have a step difference from the primary joint portion, to which the secondary membrane is attached, and a flange portion extending from the secondary joint portion for coupling with each of the insulating members, the flange portion is slidably interposed between upper plates of each of the insulating members formed of two sheets of plywood, so that each of the insulating members and the movable member are combined to be relatively slidably displaceable, and the secondary joint portion and the flange portion are formed by bending a sheet of metal, and the primary joint portion is formed by adhering a metal rod having a rectangular cross-section or a U-shaped section steel formed by bending a sheet of metal to the secondary joint portion.
11. The liquefied gas storage tank of claim 10, wherein a planar structure is disposed around the corner structure, the planar structure includes a secondary insulating panel installed on the hull structure wall and a primary insulating panel adhered to the secondary insulating panel to be adjacent to the sealing wall, the primary insulator included in the primary insulating panel and the secondary insulator included in the secondary insulating panel are formed of an insulator of the same material, and the secondary insulator has a density lower than a density of the primary insulator, and the secondary insulating panel includes one or more secondary reinforcing plates for reinforcing the secondary insulator in the secondary insulator.
12. The liquefied gas storage tank of claim 10, wherein the sealing wall includes a primary membrane in direct contact with liquefied gas and a secondary membrane installed to be spaced apart from the primary membrane by a predetermined distance, and a support plate is interposed between the primary membrane and the secondary membrane to maintain a constant interval therebetween.